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https://doi.org/10.1038/s41559-025-02692-w

Deep-pelagic ecosystems should be considered as social–ecological systems

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Deep-pelagic ecosystems are critical for climate regulation, food security and global economic activities, yet the focus of deep-sea research and conservation remains on seafloor-associated ecosystems. We discuss the overlooked deep-pelagic ecosystems and call for their recognition as social–ecological systems.

Deep-pelagic ecosystems extend from ocean depths of approximately 200 m to 5,000 m and are home to thousands of species that include fish, crustaceans, squids, salps and cnidarians. These ecosystems, which comprise over 90% of Earth's biosphere, are out of sight, and often perceived as remote and disconnected from societal concerns. But they are vital for climate regulation, food production and a variety of cultural and economic activities[1,2]. As global anthropogenic effects extend into deeper waters[3–5], a substantial shift in how society, scientists and policymakers perceive and engage with these ecosystems is urgently needed. A social–ecological systems (SES) framework offers a compelling interdisciplinary approach that views ecosystems and human societies as inseparable and interdependent, and emphasizes their dynamic and evolving interactions, which include feedback loops, adaptability and emergent properties[6]. This perspective provides a more realistic and integrated view of these ecosystems, and supports their sustainable management amid ongoing social and environmental change[6,7]. Here, we argue for the explicit recognition of deep-pelagic ecosystems as distinct, critical and irreplaceable SES.

The influence of deep-pelagic ecosystems on society

SES studies have largely focused on terrestrial and coastal marine environments. However, increasing human effects on deeper marine ecosystems and their critical links to society underscore the urgent need to expand this focus[8,9]. Deep-pelagic ecosystems not only harbour thousands of species (including some of the most abundant vertebrates), but also influence human society in multiple ways (Fig. 1). They represent a substantial share of Earth's biodiversity[1,2] and are key to nutrient cycling, oceanic layer connectivity and biogeochemical regulation through the daily vertical migration of billions of deep-pelagic organisms[8,9]. This migration facilitates the transport of 3–6 billion tonnes of carbon to the deep sea annually, a vital process for carbon sequestration and global ecosystem health[8,10]. From a carbon market perspective, this sequestration could conservatively be valued between US $300 billion and $900 billion[8]. When considering a social time preference rate of 3% (which reflects society's tendency to prioritize long-term benefits over immediate consumption), the potential economic value of this sequestered carbon could range from $10 trillion to $30 trillion[8].

Deep-pelagic ecosystems are also integral to global fisheries and provide a vital food source for major commercial species such as tuna, swordfish and Humboldt squid[1,8,9,11], which support the livelihoods and nutritional needs of millions of people worldwide. Additionally, these ecosystems sustain various charismatic and threatened species, including sea-lions, penguins and whales[11,12]. This ecological connection underpins global industries, including the global billion-dollar whale-watching sector, which generates considerable economic, educational and environmental benefits.

The fish biomass in the upper layers of the deep-pelagic waters (200–1,000 m) is estimated to be up to 19.5 gigatonnes, almost 100 times the annual yield of all global fisheries[13]. This vast biomass, alongside its rich genetic diversity, holds considerable potential for pharmaceuticals, human food and other natural product development[8,9]. It can also be argued that harvesting even a small proportion of this biomass could substantially boost fishmeal production and open up deep-pelagic ecosystems as a potential new fishery. However, the risks of exploitation are uncertain and could disrupt vital ecosystem processes[1].

Beyond its economic value, deep-pelagic ecosystems enrich human society by inspiring art, mythology, spiritual beliefs and culinary traditions[5]. For instance, the deep sea holds sacred meaning for various Indigenous Pacific cultures, and symbolizes ancestry and the origins of life. Shrouded in mystery, deep-sea species are often perceived as otherworldly and monstrous, and fuel narratives from ancient myths to modern entertainment – often exemplified by the legendary kraken of Scandinavian folklore and iconic depictions of anglerfish.

The influence of society on deep-pelagic ecosystems

Human activities are increasingly affecting deep-pelagic ecosystems: climate change, pollution and resource extraction are the primary drivers (Fig. 1). These impacts often occur simultaneously and interact in ways that exacerbate the effects on these ecosystems[3,14]. Ocean warming, for example, poses serious risks to deep-pelagic species, because most of these organisms are adapted to stable cold temperatures. Even slight increases in temperature can cause stress, which leads to changes in species distributions and alters ecological interactions[3,14]. Warming also increases ocean deoxygenation and water stratification, which limits vertical mixing, species migration and energy transfer between ocean layers. Concurrently, rising CO_2 levels are contributing to ocean acidification, which affects calcifying organisms such as pteropods (which are integral to deep-pelagic food-webs)[3,14].

The increasing prevalence of pollutants – including plastics, heavy metals, and chemical pollutants such as pesticides, herbicides and pharmaceuticals – further compounds the challenges faced by deep-pelagic species[3,4]. The escalation of deep-sea exploration worldwide also places additional stress on deep-pelagic ecosystems. Oil and

Societal benefits	Human impacts
Cultural value Mysterious and distinctive deep-pelagic species have inspired myth, literature and art for millennia	**Ocean warming** Ocean warming will stress species and alter their distributions and interactions, and increased stratification will disrupt movement and energy flow
Nutrient recycling and ecosystem conection Several species move between ocean layers and link trophic levels, and have a crucial role in nutrient recycling	**Ocean deoxygenation** Expansion of minimum oxygen zones will lead to habitat compression and control biodiversity through effects on species physiology, behaviour and interactions
Carbon sequestration Diel vertical migration has a crucial and irreplaceable role in active carbon sequestration and climate regulation	**Ocean acidification** Ocean acidification will reduce the suitable habitat for calcifying species and affect many zooplankton species that form the basis of deep-pelagic food webs
Prey resources for fishing stocks Species serve as primary prey for key commercially exploited species, such as tuna, swordfish and humboldt squid	**Pollution** Species are increasingly exposed to diverse contaminants, including fishing debris, plastics, heavy metals and chemical pollutants
Biodiversity maintance Species are an important part of marine biodiversity and support several charismatic and endangered species	**Deep-sea mining** Deep-sea extraction of oil, gas, and minerals degrades water quality, which adversely affects respiration, feeding, buoyancy and visual communication of organisms
Source of raw material Deep-pelagic biomass is a vast resource for food supplements and pharmaceuticals, and is a largely undiscovered genetic heritage	**Deep-sea fishing** Although commercial deep-pelagic fishing is not widespread, global interest is growing, and experimental projects for profitable harvesting are underway

Dynamics of social–ecological systems	
Feedback loops Human activities influence ecosystems, and ecological changes affect society	**Adaptative resilience** The system's ability to absorb disturbances, adapt to changes and maintain its core functions
Nonlinearity Minor changes can lead to disproportionate and unpredictable responses	**Complex integration** Ongoing and dynamic interactions between social and ecological elements

Fig. 1 | Key interactions between deep-pelagic ecosystems and human societies. These interactions highlight the mutual influences of deep-pelagic ecosystems and human societies, and the inherent dynamics of SES.

gas extraction can markedly alter the physicochemical conditions of the water column, particularly in the event of spills or accidents[3]. Similarly, sediment plumes and noise pollution from seabed mineral exploration may have pronounced ecological effects on deep-pelagic species[15].

An SES approach to managing human impacts

Deep-pelagic ecosystems are closely linked to at least six of the seventeen United Nations Sustainable Development Goals (SDGs). These include SDG 14, which focuses specifically on marine biodiversity and its sustainable use, as well as food security (SDG 2), health and well-being (SDG 3), economic growth (SDG 8), climate action (SDG 13), and global partnerships for sustainable development (SDG 17).

Despite these strong connections, deep-pelagic ecosystems remain largely overlooked and human impacts are advancing faster than our ability to understand their consequences and develop effective management measures. Although global efforts to reduce CO_2 emissions and enhance carbon sequestration are advancing, the vital role of deep-pelagic ecosystems in transporting billions of tonnes of carbon annually at no economic cost remains poorly recognized. Without greater awareness, these ecosystems will continue to be undervalued in climate policies and conservation initiatives.

The recent High Seas Treaty marks an important step towards the conservation of marine biodiversity beyond national jurisdictions, and establishes frameworks for protection, sustainable use and equitable resource sharing. However, as long as deep-pelagic ecosystems remain unexplored and undervalued, they will remain overlooked or oversimplified in discussions and policy development under this treaty. For example, global initiatives such as the Blue Carbon framework — which focuses on enhancing carbon sequestration through marine habitat conservation — tend to prioritize well-known coastal ecosystems and single end points (for example, mangroves), but rarely consider deeper, less-visible processes that are critical to global carbon cycling. This lack of integration means that deep-pelagic ecosystems are frequently omitted from global carbon assessments and rarely mentioned in blueprints presented by major international bodies.

The limited incorporation of deep-pelagic ecosystems into global frameworks is largely due to knowledge gaps. It has only recently become clear that, despite their complexity and our limited understanding, deep-pelagic ecosystems contain ten times more biomass and exert a far greater influence on global ecosystem processes than previously thought[10,13]. Furthermore, the vast size of the deep ocean is often invoked to justify its marginalization in conservation priorities compared to coastal ecosystems. However, this perspective assumes that deep-pelagic species are globally redundant. Recent evidence suggests otherwise: these species are highly specialized and heterogeneous, and have irreplaceable roles in several ecosystem processes[1]. The distribution of life in the deep ocean is also uneven; it contains diverse communities, biodiversity hotspots and areas that contribute disproportionately to global ecosystem processes[1,2].

BOX 1

Benefits of a SES approach to deep-pelagic ecosystems

System-level understanding

The SES framework integrates multiple disciplines to uncover complex feedback loops and emergent properties that are often missed by current discipline-specific approaches. By using SES tools such as system dynamics modelling, governance matrices and participatory scenario analysis, the framework supports more accurate predictions of ecosystem responses, policy effects and socioeconomic trade-offs. Moreover, it fosters knowledge coproduction among scientists, policymakers and stakeholders, which promotes a more holistic and actionable understanding of deep-pelagic ecosystems. This collaborative approach is key to addressing the gaps in current research and management.

Adaptive and resilient management

The SES approach fosters continuous learning and stakeholder engagement, and embraces uncertainty and dynamic feedback loops — which are inherent in deep-pelagic ecosystems and often underestimated in traditional management frameworks. By considering potential regime shifts, ecological tipping points and socioeconomic disruptions, SES-informed strategies are inherently more flexible and responsive than static policy approaches. This perspective ensures that both ecological and social resilience are supported, which enables management approaches to evolve alongside new scientific knowledge and environmental changes, and ultimately enhances our ability to cope with unforeseen challenges.

Multiscale and inclusive governance

The global nature of deep-pelagic ecosystems poses unique governance challenges, as they transcend national jurisdictions and involve diverse actors with different interests. The SES framework offers a valuable structure for integrating local, regional and international knowledge systems, and helps to bridge the gap between ecological processes and governance structures. By fostering inclusive decision-making, stakeholder collaboration and cross-sectoral coordination, SES-informed governance can enhance policy legitimacy, enforcement effectiveness and long-term sustainability. These qualities are crucial for ensuring that deep-pelagic ecosystems receive appropriate attention and action in global conservation and resource management efforts.

We argue that considering deep-pelagic ecosystems as SES is crucial to addressing these challenges and ensuring their resilience in the face of environmental change (Box 1). Explicitly addressing the interdependencies within SES frameworks enhances our ability to account for the resilience, adaptability and intricate dynamics between society and ecosystems — critical elements for achieving sustainability[6,7]. This includes, for example, recognizing that these dynamics exhibit nonlinear feedbacks (in which interactions can trigger sudden, sometimes-irreversible shifts) and are characterized by heterogeneity, which enables different components to adapt uniquely to changes[6] (Fig. 1). This approach also recognizes that ecological and societal processes are deeply intertwined and operate across multiple spatiotemporal scales. Fast and slow dynamics influence one another: ecological changes shape human responses and societal actions, in turn, drive environmental shifts. These interactions can either stabilize or destabilize the system, which sometimes leads to abrupt and irreversible changes. Integrating SES principles into research and policy development can therefore help to improve adaptive management strategies, and ensure that deep-pelagic ecosystems are protected in a way that reflects their complexity and importance[6,7] (Box 1).

Although no formal designation of SES exists, their adoption depends on how different disciplines prioritize and integrate them. From a scientific perspective, advancing this concept requires incorporating more human dimensions into deep-pelagic ecosystem studies, fostering interdisciplinary collaboration, and securing targeted funding to support SES-focused initiatives[7]. From an economic perspective, deep-pelagic SES should be integrated into blue economy strategies by recognizing their role in supporting fisheries, tourism, carbon transport and biotechnology, while assessing potential risks from human activities. At the policy level, their role in carbon cycling should be emphasized in global climate discussions, such as the Conference of the Parties (COP), to encourage international collaboration in research and conservation. Deep-pelagic SES must also be explicitly considered in policy frameworks developed under agreements such as the High Seas Treaty and initiatives such as the Blue Carbon framework. Rather than viewing the current knowledge gap as a justification for inaction, this gap should catalyse precautionary measures and further research. Acknowledging deep-pelagic SES as essential and interdependent with human societies could increase funding opportunities, improve governance structures, and promote their inclusion into local and global policy frameworks. Given the absence of a dedicated deep-pelagic governance body (unlike the International Seabed Authority), science has a de facto governance role by shaping the way that these SES are understood. We, members of the scientific community, must therefore take the lead in actively integrating deep-pelagic SES into research agendas, conservation priorities and policy discussions. Deep-sea SES must be not only subjects of study and curiosity but also central to shaping global policies and practices.

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Published online: 18 April 2025

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Acknowledgements
This research represents a collaborative effort within the framework of the IJL TAPIOCA (www.tapioca.ird.fr). We gratefully acknowledge P. Guillotreau, who has considerably enhanced this work.

Competing interests
The authors declare no competing interests.

Additional information
Peer review information *Nature Ecology & Evolution* thanks the anonymous reviewers for their contribution to the peer review of this work.